Money Laundering Is Not an Outlier: Detecting Illicit Bitcoin Transactions Without Illicit Labels

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Abstract

Cryptocurrency exchanges must flag laundering transactions, but confirmed illicit labels are scarce and late, while licit ones are cheap. On the Elliptic Bitcoin graph (203,769 transactions, 49 time steps) we show that illicit transactions are *not* outliers: only 1.7% exceed a 3σ rule versus 22.3% of licit ones, so one-class detectors rank them *below* normal (test ROC-AUC 0.17). We therefore train a positive-unlabeled detector (LvU) that separates licit from unlabeled transactions and never sees an illicit label. On a chronological test split, LvU reaches PR-AUC 0.546 ± 0.024 ($8.4\times$ prevalence) and 82% precision in its top 100 alerts, matching a supervised GraphSAGE network and recovering 69% of supervised XGBoost's PR-AUC. Ablations show that 1% illicit contamination of the licit labels destroys this gain, and every method fails after a dark-market shutdown.

1 Introduction

Scenario and stakeholders. Virtual-asset service providers must screen the transactions they touch for money laundering. Every alert goes to a compliance analyst, who decides whether to freeze funds and file a suspicious-activity report. A missed laundering transaction (FN) exposes the provider to regulatory penalties; a false alert (FP) costs analyst time from a fixed weekly review budget.

Motivation. Supervised detectors work well on Bitcoin data (Weber et al., 2019) but need confirmed illicit labels, which arrive late, whereas *normal* labels are cheap (known exchanges are tagged automatically). One-class models, which treat anomalies as rare points far from normal data (Chandola et al., 2009), are the natural label-free choice, yet Lorenz et al. (2020) found that they perform poorly and that illicit instances are not outlying. We quantify this failure, explain it, and turn the explanation into a detector.

Definition of anomaly. Our data belong to the **Graph** category: the unit of detection is one Bitcoin transaction, a node in a payment graph. A transaction is anomalous if the Elliptic labels attribute it to an illicit entity (scams, malware, ransomware, Ponzi schemes, terrorist organisations); it is normal if attributed to a licit entity (exchanges, wallet providers, miners). Example normal sample: tx 378120377 (step 16, licit; `local_53`$=0.92$, in/out degree 0/2). Example anomaly: tx 232014511 (step 1, illicit; `local_53`$=-0.49$, degree 1/1).

Research question. *Can a detector fitted without any illicit label rank illicit Bitcoin transactions above licit ones on future time steps, even though illicit transactions are not outliers?*

Contributions. (C1) Illicit transactions are less extreme than licit ones (1.7% vs. 22.3% beyond 3σ), which inverts one-class detectors (ROC-AUC 0.17–0.20; §4, §7). (C2) A licit-vs-unlabeled (LvU) detector reaches test PR-AUC 0.546 and P@100 0.82 without illicit labels: far above label-free baselines, on par with a supervised graph neural network, below supervised trees (§7). (C3) It is robust to the unlabeled pool size but collapses when 1% of licit labels are wrong (§7.1). (C4) 42% of its misses fall after a dark-market shutdown, a blind spot shared with supervised models (§8).

2 Related Work

Data category and literature. Our data are in the **Graph** category (transactions are nodes, payment flows directed edges); its key challenge is multi-hop structure, as laundering routes funds through chains of transactions. Weber et al. (2019) released Elliptic and compared supervised models on a 70:30 temporal split (Table 1); Random Forest on all features beat graph convolutional networks. EvolveGCN (Pareja et al., 2020) models the graph

Work	Data	Paradigm	Method	F1
Weber et al. (2019)	Elliptic	sup.	RF, AF	0.788
Weber et al. (2019)	Elliptic	sup.	GCN	0.628
Weber et al. (2019)	Elliptic	sup.	EvolveGCN	0.720
Lorenz et al. (2020)	Elliptic	unsup.	LOF (best)	$\leq$0.19
Lorenz et al. (2020)	Elliptic	unsup.	IF	$\leq$0.01
Ours	Elliptic	PU	LvU	0.398
Ours	Elliptic	sup.	RF, AF	0.541

Table 1: Prior results (illicit-class F1; sup./unsup. = (un)supervised). Prior work trains on steps 1–34; we train on 1–29 and tune τ on 30–34, so numbers give context, not a direct comparison.

sequence over time (illicit F1 0.720 in Weber et al., 2019) and reports that no model recovers after the dark-market shutdown. Lorenz et al. (2020) found that seven unsupervised detectors reach illicit F1 of at most 0.19 (LOF), and proposed active learning, which matches a supervised baseline with 5% of the labels. Positive-unlabeled (PU) learning trains a classifier from one labeled class plus unlabeled data (Elkan and Noto, 2008; Bekker and Davis, 2020); Deep SAD (Ruff et al., 2020) instead uses a few labeled anomalies.

Positioning and method selection. Tier 1 assumes anomalies are extreme in some feature (3σ rule) or nothing at all (random score). Tier 2 assumes illicit mass is over-represented in unlabeled relative to licit data (our LvU); we also report the one-class Isolation Forest (Liu et al., 2008), which assumes anomalies are easy to isolate. Tier 3 uses full supervision: Random Forest (Breiman, 2001) as in Weber et al. (2019), XGBoost (Chen and Guestrin, 2016), and a graph neural network. Graph context enters in two ways: as one-/two-hop aggregates and degree (A4), and through message passing with GCN (Kipf and Welling, 2017) or GraphSAGE (Hamilton et al., 2017) layers, used both as a supervised reference (T3c) and as the classifier inside LvU (T2″). Unlike prior work, we fit without illicit labels, explain *why* outlier detectors invert, and analyse failures per transaction.

3 Task Definition and Data

Data set. We use the Elliptic Data Set (Weber et al., 2019; Elliptic, 2019) (CC BY-NC-ND 4.0; an unmodified copy is in our repository). It contains $N = 203{,}769$ transactions and 234,355 edges in 49 time steps about two weeks apart; edges never cross time steps. Each node has $D = 166$ anonymised features: the time step, 93 local features (LF; e.g. fees, input/output counts and volumes) and 72 aggregates over one-hop neighbours (AF). From the edge list we add in/out degree and two-hop aggregates (mean of the neighbours' AF). §1 defines the unit of detection and gives a normal and an anomalous example.

Split	Steps	Tx	Licit	Illicit	Unknown	Ill./lab.
train	1–29	120,804	23,510	2,871	94,423	10.9%
val	30–34	15,461	2,922	591	11,948	16.8%
test	35–49	67,504	15,587	1,083	50,834	6.5%
all	1–49	203,769	42,019	4,545	157,205	9.8%

Table 2: Chronological split shared by all methods.

Labels. Ground truth comes from a heuristics-based labelling of the entities behind each transaction (Weber et al., 2019): 4,545 illicit, 42,019 licit, and 157,205 *unknown*. Labels may be noisy (heuristic, entity-level attribution); unknown transactions contain both classes and are never evaluated, but LvU uses them as unlabeled data. The paradigm is semi-supervised: licit and unlabeled data for fitting, illicit labels only for validation-based selection.

Class imbalance. Overall, 2.2% of transactions are illicit; among labeled ones the ratio is 9.8% vs. 90.2% (1 : 9.2), and on the test split 6.5% (1 : 14.4). A model that predicts "always licit" reaches 93.5% test accuracy with zero recall, so we never report accuracy. We use threshold-free ranking metrics (PR-AUC) and choose τ to maximise F1 rather than relying on a 0.5 cut-off.

4 Exploratory Feature Analysis

All statistics use the training split only (Figure 1).

Distribution contrast. Figure 1a shows `local_53`, the feature with the largest Kolmogorov–Smirnov (KS) distance between classes (0.66, mutual information 0.21 nats). Most illicit transactions sit on a single value (-0.49, shared by 83% of illicit but 18% of licit training transactions), whereas licit values spread over the whole range. Figure 1b is the key observation: only 1.7% of illicit transactions have any local feature beyond 3σ of the licit distribution, against 22.3% of licit transactions. In PCA space (Figure 1c), illicit points occupy one compact region; licit points scatter over many clusters.

Separability and redundancy. Local features separate the classes better than one-hop aggregates

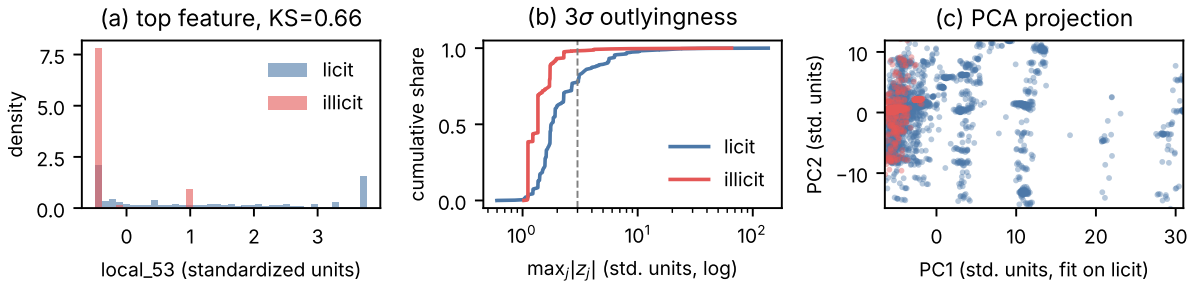


Figure 1: Training split only. (a) The most separating feature: illicit transactions concentrate on one value. (b) Cumulative share of the largest $|z|$-score (w.r.t. licit statistics): illicit transactions are *less* extreme than licit ones; dashed line = 3σ. (c) PCA fitted on licit data: illicit points form a compact cluster inside the region occupied by licit ones.

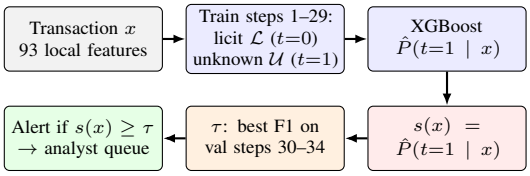


Figure 2: LvU pipeline. No illicit label is used for fitting; labeled illicit transactions of the validation steps only set τ.

(mean KS 0.38 vs. 0.27; 63 vs. 30 features with KS > 0.3; mean MI 0.088 vs. 0.060 nats), degree (0.20) and two-hop aggregates (0.18). 89 feature pairs have $|r| > 0.95$, and licit behaviour drifts moderately within training (mean KS 0.11, steps 1–14 vs. 15–29). On training edges, 56% of the labeled neighbours of illicit transactions are illicit (5× the base rate), but 59% of all their neighbours are unknown (licit: 46%).

Hypotheses. **H1**: Outlier-based scores (3σ, Isolation Forest, Mahalanobis) will rank illicit transactions *below* licit ones. **H2**: A model that contrasts licit data with unlabeled data will rank them above, because the compact illicit mode is denser in the unlabeled pool than among labeled licit transactions. **H3**: Graph context (aggregates, degree, message passing) will add little, as unknown neighbours dominate. We test H1–H2 in §7–7.1 and H3 in §7.1.

5 Method

Pipeline. Figure 2 shows the Tier 2 pipeline. The input is the 93 local features of a transaction, used as released (already scaled); the time step is dropped so that the model cannot memorise periods. We build a training set from the training steps: every labeled licit transaction gets target $t = 0$ and every unknown transaction gets $t = 1$ (23,510 vs. 94,423 rows); illicit training rows are *not* used. An XGBoost classifier (300 trees, depth 6, learning rate 0.1, row/column subsampling 0.8) learns $\hat{P}(t{=}1 \mid x)$.

Anomaly score. The score is $s(x) = \hat{P}(t{=}1 \mid x)$. For a well-calibrated classifier, s is monotone in the density ratio $p_{\mathcal{U}}(x)/p_{\mathcal{L}}(x)$. If the unknown pool is a mixture $p_{\mathcal{U}} = \pi\, p_{\text{ill}} + (1-\pi)\, p_{\text{lic}}$ and labeled licit data follow p_{lic}, the ratio equals $(1-\pi)+\pi\, p_{\text{ill}}(x)/p_{\text{lic}}(x)$ and therefore ranks transactions exactly by how much more likely they are under the illicit than the licit distribution (Elkan and Noto, 2008). Unlike an outlier score, this ranking rewards a dense illicit mode that licit data rarely reach (H2).

Threshold and error costs. τ is the score that maximises F1 on the labeled transactions of validation steps 30–34; the test split is never used. An FN lets laundered funds pass (regulatory risk); an FP costs analyst minutes. Because an FN is costlier, we accept the recall-leaning τ that F1 selects: about 2.5 false alerts per caught transaction at 65% recall (alternatives in §7.1).

6 Experimental Setup

Split and leakage prevention. All methods share one chronological split by time step (Table 2): train 1–29, validation 30–34, test 35–49, i.e. 59/10/31% of steps and 57/8/36% of labeled transactions. One-class models fit licit training rows only; LvU fits licit and unknown training rows; supervised models fit labeled training rows. Scalers (3σ, LOF, kNN) are fit on licit training rows; the test set is scored once per seed.

Configurations. LvU: feature set $\in$ {LF, AF, AF+degree}, depth $\in \{4, 6, 8\}$ and trees $\in$

$\{100, 300\}$, chosen by validation PR-AUC (LF, 6, 300; 0.901). Isolation Forest: 200 trees, `max_samples` $\in \{256, 2048\}$ (2048, LF). Random Forest as in Weber et al. (2019) (50 trees, 50 features per split, AF); XGBoost as LvU, on AF. GNNs (PyTorch Geometric, Fey and Lenssen, 2019): two GCN or GraphSAGE layers, full-batch (edges never cross steps), Adam (learning rate 0.01), early stopping on validation PR-AUC ($\leq$1000 epochs, patience 100); validation picked GraphSAGE, width 64 (of 64/128) and AF for both roles. Stochastic methods use 5 seeds; software: Python 3.13, scikit-learn 1.9 (Pedregosa et al., 2011), XGBoost 3.4, PyTorch 2.14; hardware: 4 vCPUs, 16 GB RAM, no GPU.

Metrics. (1) **PR-AUC** (average precision) is our primary metric: it ignores the large number of easy true negatives and is informative under imbalance (Saito and Rehmsmeier, 2015); its random baseline equals the prevalence. (2) **P@100**, the precision of the 100 highest-scored test transactions, models a fixed analyst budget. (3) **F1@τ** evaluates the deployed decision. ROC-AUC is reported to show inversion (< 0.5).

7 Results

Main comparison. Table 3 supports H1: the 3σ rule and the one-class Isolation Forest rank illicit transactions below licit ones (ROC-AUC 0.33 and 0.17) and fall under the random baseline in PR-AUC. Their val-tuned τ degenerates to "alert on everything" (F1 0.12, as for random scores). LvU supports H2: PR-AUC rises to 0.546 ± 0.024 (8.4$\times$ prevalence) and 82% of its top-100 alerts are illicit. The gap to the best label-free baseline (T1a) is significant (paired t-test over seeds, $p < 10^{-5}$; bootstrap 95% CI of the PR-AUC gap [0.450, 0.513]).

Against Tier 3. Supervised trees remain clearly better (XGBoost 0.790, RF 0.767; bootstrap CI of the gap vs. XGBoost $[-0.268, -0.218]$, $p < 10^{-4}$), though LvU recovers 69% of XGBoost's PR-AUC without illicit labels. It loses because it learns "unlabeled-like" rather than "illicit": licit transactions resembling the unlabeled mass also score high (§8). RF's F1 varies across seeds (0.42–0.80) while its PR-AUC does not: a τ tuned at 16.8% validation prevalence transfers poorly to 6.5% at test. Trees train in under 6 s and GNNs in 2–5 min; all score in 1–12 μs per transaction.

Message passing (Graph-specific). The supervised GraphSAGE (T3c) reaches PR-AUC 0.552 ± 0.035, statistically indistinguishable from label-free LvU ($p = 0.79$; CI $[-0.038, 0.027]$) and 0.24 below XGBoost, consistent with trees beating GNNs on Elliptic (Weber et al., 2019). Inside LvU, message passing fails (T2″, 0.189): averaging over neighbourhoods dominated by unknown transactions makes most nodes look "unlabeled" and erases the licit/unknown contrast (H3).

7.1 Controlled Ablations

A1: contamination of the normal labels (AD-specific). Replacing just 1% of the licit training rows with illicit ones (235 rows) cuts PR-AUC from 0.546 to 0.142, far beyond the seed std. Because illicit transactions form one compact mode, a few hundred of them labeled $t = 0$ suffice to tell the classifier that this mode is "licit". This is an honest negative result: LvU needs clean licit labels.

A2: size of the unlabeled pool (AD-specific). 5% of the unknown pool (4.7k rows) gives the same PR-AUC (0.546); differences stay within one std, so a small unlabeled sample already holds the illicit mode.

A3: scoring function (AD-specific). Holding features and data fixed, outlier scores invert (Isolation Forest 0.037, Mahalanobis with Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004) 0.038). Local density scores, LOF (Breunig et al., 2000) and a kNN density ratio between licit and unlabeled data, recover some signal (0.19). The classifier-based density ratio is best (0.546).

A4: graph features (Graph-specific). One-hop aggregates ($+0.011$), degree (-0.037) and two-hop aggregates (-0.010) all change PR-AUC by less than one seed std, which supports H3. They raise recall but lower F1, because the validation threshold transfers worse.

Threshold strategy. With the same scores, the label-free rule τ = 95th percentile of licit training scores doubles the alerts to 5,420 (precision 0.16, recall 0.80, F1 0.27); 30 alerts per step (450 in total) give the best precision, 0.48, at recall 0.20 (std $\leq$0.01).

8 Analysis

Temporal failure. Figure 3 reproduces the observation of Weber et al. (2019): after a dark market

Tier	Method	PR-AUC	ROC-AUC	F1@τ	P@100	PR-AUC by period steps 35–42	PR-AUC by period 43–49	Fit (s)	Score (μs/tx)
T1a	Random score	0.065±0.001	0.495±0.006	0.122±0.000	0.060±0.017	0.091	0.027	0.1	0.0
T1b	3σ rule (max $\lvert z\rvert$)	0.053	0.326	0.010	0.040	0.075	0.028	0.1	0.2
T2′	One-class Isolation Forest	0.038±0.000	0.170±0.008	0.121±0.002	0.000	0.053	0.021	0.7	11.9
T2″	LvU-GraphSAGE	0.189±0.017	0.799±0.018	0.279±0.023	0.190±0.046	0.289	0.036	120	1.6
T2	**LvU classifier (ours)**	0.546±0.024	0.865±0.002	0.398±0.011	0.822±0.020	0.743	0.025	4.6	1.5
T3a	Random Forest (sup.)	0.767±0.003	0.891±0.004	0.541±0.138	**1.000**	0.897	0.028	5.7	2.2
T3b	XGBoost (sup.)	**0.790±0.001**	**0.922±0.002**	**0.675±0.056**	**1.000**	**0.914**	0.036	2.8	1.7
T3c	GraphSAGE (sup.)	0.552±0.035	0.874±0.003	0.568±0.028	0.842±0.086	0.687	0.029	291	1.6
prevalence (= random PR-AUC)		0.065				0.092	0.025		

Table 3: Test results (labeled transactions of steps 35–49), mean ± std over 5 seeds (deterministic methods: one run). F1 uses τ tuned on validation (3σ rule: $\tau = 3$). Best per column in bold. Period columns split the test set at the dark-market shutdown (step 43).

Setting	PR-AUC	Δ	F1@τ
A1: illicit rows hidden in licit side			
0% (ours)	0.546±0.024	—	0.398
1%	0.142±0.007	−0.404	0.246
5%	0.098±0.003	−0.448	0.207
10%	0.091±0.004	−0.455	0.196
A2: fraction of unlabeled pool used			
5%	0.546±0.018	+0.000	0.406
25%	0.520±0.047	−0.026	0.425
50%	0.551±0.040	+0.005	0.418
100% (ours)	0.546±0.024	—	0.398
A3: label-free scoring function			
one-class IF	0.037±0.000	−0.509	0.122
one-class Mahalanobis	0.038	−0.508	0.122
one-class LOF	0.190	−0.356	0.139
PU kNN ratio	0.194	−0.352	0.188
PU classifier (ours)	0.546±0.024	—	0.398
A4: feature set			
LF, 93 local (ours)	0.546±0.024	—	0.398
AF (+72 aggregates)	0.557±0.037	+0.011	0.296
AF + in/out degree	0.509±0.043	−0.037	0.316
AF + 2-hop aggregates	0.536±0.046	−0.010	0.351

Table 4: Ablations of LvU on the test split, one factor at a time (5 seeds unless deterministic; F1: mean). Δ = PR-AUC change vs. the full method.

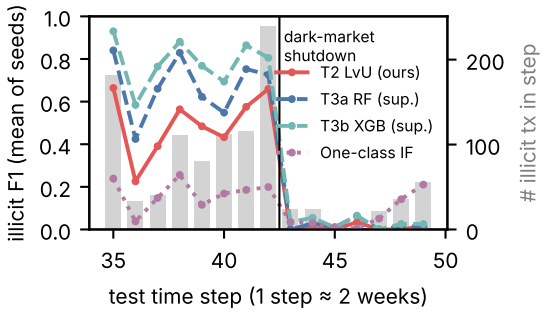


Figure 3: Illicit F1 per test step (lines, left axis) and number of illicit transactions (bars, right axis). Every method collapses after the dark-market shutdown at step 43. The one-class IF rises only because it alerts on everything.

closes at step 43, every model fails. LvU's F1 drops from 0.50 (steps 35–42) to 0.01 (43–49), and its PR-AUC from 0.743 to 0.025, which equals the prevalence (Table 3). Only 15.6% of test illicit transactions fall in steps 43–49, but they account for 42% of the misses of LvU. Of the 398 misses (seed 0, $\tau = 0.970$), 69% are also missed by supervised RF: the blind spot lies in the data, not in the missing labels.

Failure cases. Table 5 lists four test errors. #1 is a complete inversion: an illicit transaction with $s = 0.014$ whose `local_53`= 1.81 lies above 99.6% of illicit but at the 63rd percentile of licit training transactions; RF misses it too (0.04). Cases #2 (FN) and #4 (FP) get the identical score 0.970, just at τ: they share 73 of 93 local features (L2 distance 4.1 vs. a median of 10.2 to licit transactions) yet carry opposite labels, so no threshold separates them (label noise, or entities reusing the same wallet software). #3 is a confident FP ($s = 1.000$): a licit transaction on the illicit mode `local_53`= −0.49 that RF correctly scores 0.03; without illicit labels, LvU cannot tell which unlabeled mass is illicit. **Shared pattern**: errors come from overlapping modes and post-shutdown drift, not model capacity; the first fix we would try is a few analyst-confirmed labels from the new period (active learning, Lorenz et al., 2020), not a larger model.

Interpretability. We explain LvU with exact TreeSHAP (Lundberg et al., 2020). Globally (mean |SHAP| over test illicit transactions), `local_53` (0.88 log-odds), `local_47` and `local_76` (0.56 each) dominate; `local_53` is also the top EDA feature, so the model uses the illicit mode of §4. For the flagged tx 162428632 (Figure 4;

#	Type	txId	Step	$s(x)$	Top-SHAP feature	RF	Likely cause
1	FN	97069571	40	0.014	local_48= 0.09	0.04	licit-typical profile
2	FN	73003591	42	0.970	local_53= −0.49	0.76	73/93 features equal #4
3	FP	169360875	45	1.000	local_53= −0.49	0.03	licit tx in illicit mode
4	FP	73160604	42	0.970	local_53= −0.49	0.41	73/93 features equal #2

Table 5: Failure cases of LvU (seed 0, $\tau = 0.970$). RF = supervised Random Forest score (mean of seeds; its mean $\tau \approx 0.37$). Top-SHAP = local feature with the largest attribution for this transaction.

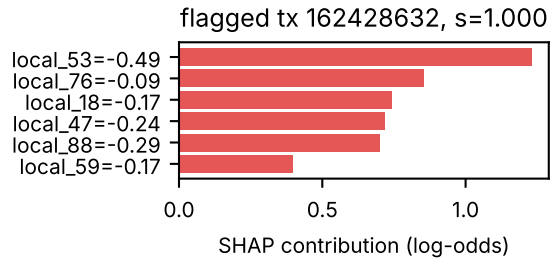


Figure 4: TreeSHAP attributions (log-odds of $t = 1$) for the highest-scored true positive of LvU (seed 0): its value on the illicit mode `local_53`$= -0.49$ contributes most.

$s = 1.000$, RF 0.94), `local_53`$= -0.49$ adds 1.23 log-odds and five other features 0.4–0.9 each. An analyst sees a pattern common in illicit flows: a reason to review, not proof.

Practical impact. At the validation threshold, LvU raises 2,459 alerts over 15 steps (≈164 per two-week step) and catches 65% (≈704) of the 1,083 illicit test transactions, with 71% false alerts; under a budget of 30 alerts per step, 48% of alerts are illicit, 7.4× the yield of random review. Supervised RF catches ≈100 more with fewer alerts, so LvU is a cold-start triage tool for providers without confirmed illicit cases.

9 Conclusion and Limitations

Key findings. Yes, with qualifications: a detector fitted only on licit and unlabeled transactions ranks illicit Bitcoin transactions far above chance on future steps (PR-AUC 0.546 vs. 0.065; P@100 0.82), whereas outlier detectors invert (ROC-AUC 0.17), because illicit transactions form a compact, non-extreme mode. It matches a supervised GraphSAGE and reaches 69% of supervised XGBoost's PR-AUC, but it fails after the dark-market shutdown, like every model we tested.

Reflection and future work. The decision that mattered most was testing outlyingness in EDA (Figure 1b); starting again, we would do it before implementing any one-class model. Next steps: (i) query analyst labels for the highest-scoring unlabeled transactions after a drift alarm, to address the post-shutdown misses; (ii) estimate the class prior π of the unlabeled pool (Bekker and Davis, 2020) to calibrate τ across periods.

Limitations. *Data coverage*: one blockchain and 49 two-week steps (undisclosed calendar dates), with entity-level heuristic labels and anonymised features; 77% of transactions are unlabeled, so all scores refer to labeled ones only. *Assumptions*: LvU assumes clean licit labels (A1 shows 1% noise is fatal) and that the unlabeled pool contains the illicit modes of the future, which fails after the shutdown. *Graph*: multi-hop structure enters as aggregates and two-layer message passing; deeper or temporal GNNs (e.g. EvolveGCN) are untested. *Generalisation*: the inversion may not hold for genuinely extreme fraud (e.g. large one-off thefts).

Deployment considerations. *Concept drift*: illicit behaviour changed abruptly at step 43; retrain every few steps and monitor with a small labeled audit sample, since scores alone do not reveal a collapse. *Latency and privacy*: scoring takes ≈1.5 μs per transaction, far below block times; blockchain data are public, but linking addresses to people falls under data-protection and AML rules.

Contributions of Each Member

- Member A: EDA; vetted the topic.
- Member B: assessed candidate topics.
- Member C: presentation slides.
- Member D: drafted the paper.
- Member E: code, slides, coordination.
- Member F: coordination, review.
- Member G: review, topic vetting.
- Member H: obtained the dataset.

AI usage statement. We used Claude Code for code and drafting; the team chose the question and verified every number and citation.

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